

SAEINDIA

SOUTHERN SECTION

Bicycle Design Challenge 2025-26



Design Report - 1

Team Name	Pegasus
Team ID	BDC2025020

NAME	TEAM COMPOSITION
Bhuvanan M	Captain
Kusumarekha M	Vice captain
Dharaneshwaran B	Designer-1
Ambika B	Designer-2
Aravinth R	Driver
Keerthana M	Co-Driver
Dharanidharan T	Research and development

FACULTY MENTOR:

ABSTRACT:

- This project presents a light, small, and environmentally friendly bicycle that aims to address the demands of contemporary urban commuters. At the centre of the design is an innovative Z-frame structure, which offers better strength, balance, and visual impact while keeping a minimal footprint. For maintaining comfort for the rider on uneven surfaces, a centre shocker suspension system is incorporated into the frame. The bicycle also has a regenerative braking system in conjunction with conventional mechanical brakes, providing both efficient energy and sure stopping power.
- More user-friendly design features include a horizontal adjustable and a handlebar with improved grip for more control. Platform pedals give the rider a large, stable platform to enhance foot support and fatigue reduction. Standard wheels for consistent riding, and easy parking support for everyday practicality. Made from environmentally friendly materials, the bicycle is conducive to green commuting and a practical, comfortable, and ethical solution for daily urban transport.

INTRODUCTION:

As cities become increasingly crowded, there is an increasing demand for environmentally friendly, lightweight, and intelligent transportation means. The bicycle is a perfect solution, but contemporary use requires more than the frame and wheels. This project seeks to create a bicycle that provides comfort, security, and sustainability with practical features suitable for everyday urban commuting. With a Z-frame, middle suspension, and specially designed components, the aim is to redefine what a city bike can be robust, compact, and ecologically sound.

SCOPE & OVERVIEW OF THE PROJECT

SCOPE:

- **Design Emphasis:** Practicability, structural integrity, and urban versatility.
- **Materials:** smooth and recyclable materials.
- **User Experience:** A spring system with support at the joints positioned beneath the seat offers inbuilt cushioning that supports comfort over rough ground without height adjustments.
- **Sustainability:** Minimal carbon footprint with regenerative aspects and recyclable materials.
- **Urban Utility:** Easy parking, strong construction, adaptable to different road conditions.

- This project proposes a light and compact bicycle tailored for contemporary metropolitan urban travel. It has a strong, structurally efficient Z-frame with enhanced balance, ruggedness, and streamlining. Standard mechanical brakes are used on the bicycle to provide dependable and secure stopping performance under typical daily conditions. One important comfort feature is the spring mechanism located underneath the seat that cushions shocks from riding and renders height adjustment unnecessary.
- The handlebar is ergonomically styled to accommodate a natural riding position, which improves control and rider comfort. Platform pedals offer a spacious, stable platform for efficient pedalling, and a side-mounted kickstand allows for easy and rapid parking. Constructed from environmentally friendly materials, the bike encourages sustainable commuting. The overall aesthetics focus on the comfort of the user, functional usability, and greenness.

PROJECT EXECUTION PLAN

Week 1-3:

- Concept Development
- Design Development
- Geometry & Calculation

Week 3-4:

- CAF
- FEA modification

Week 5-6:

- Prototype/Tooling/Raw material/manufacturing process
- Modification based on prototype testing
- Design review/initial manufacturing

Week 6-7:

- BOM/Cost
- Estimation/Manufacturing

Week 8-9:

- Final project review and display
- Catalogue and USP of the project

Week 10-11:

- For the total completion of the project.

CONCEPTUALISATION:

Frame:

The bike embraces a radical and contemporary Z-frame design, pushing aside classic triangular configurations in favour of more space-efficient use. The structure boasts enhanced strength and carries its weight more evenly around the frame. The distinctive shape improves balance, minimises frame stress, and facilitates urban use in confined areas. It is particularly useful for light constructions, helping the bike stay portable and responsive. Built from environmentally friendly metals such as aluminium or reusable steel, the frame is also sustainable. Compact in design, it is simple to store or haul, particularly in crowded city centres. Z-frame also enables a clear visual identity, which differentiates it from standard bicycles. Designated mount points on the frame provide optimal space for accessories such as bottle cages or baskets. Attention to frame welding and reinforcement of the joints provides durability for normal urban life. The design is not just functional, but also attractive, with an appeal to both enthusiasts and commuters.

Comfort & Ergonomics

For added rider comfort, a spring suspension unit is installed directly under the seat, with jointed mounts. Such a spring dampens road shock and vibrations on rough or bumpy roads, which enhances ride quality. This design offers automatic suspension comfort without manual adjustment.

The handlebar features an ergonomic angle to enable a comfortable upright posture for minimising fatigue. It's textured, anti-slip grips enhance wet and bumpy ride control. Handlebar width and reach are calculated on the basis of average human body proportions for easy, relaxed riding. Seat, pedal, and handlebar positions have geometry based on ergonomic principles of comfort for extended periods. All these components make the bicycle perfect for a broad range of rider body types and heights.

Comfort is paid attention to, making the ride comfortable for everyday commuting, minimising joint and muscle stress. Working together, the seat and handlebar system form a rider-first experience based on posture and stability.

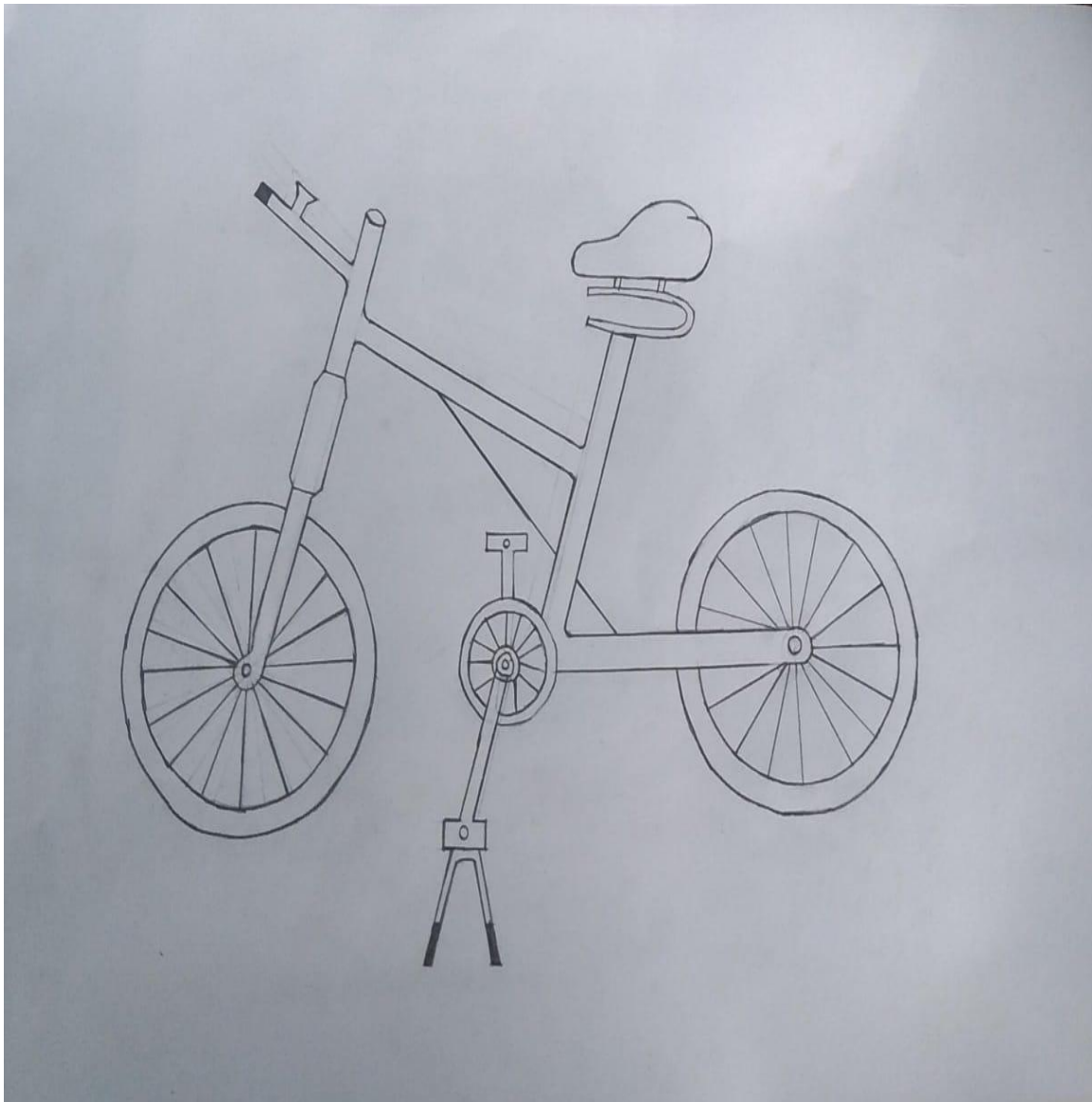
Functional components:

The bike is equipped with standard mechanical brakes that offer consistent and predictable stopping performance. Brake levers are strategically positioned for convenient reach and rapid response, enhancing safety in traffic or downhill routes. Platform pedals are provided, with a large, flat, anti-slip area for accommodating all types of shoes. The pedals enable improved foot control and power transfer, perfect for stop-and-go urban riding.

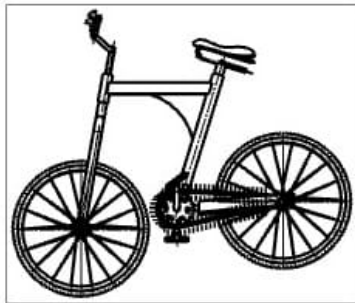
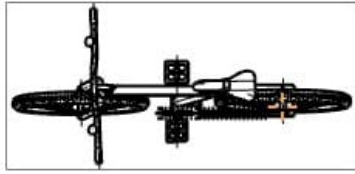
A kickstand mounted on the side gives firm and convenient parking, allowing the bicycle to be used on rough terrain. Wheels are standard size, with puncture-resistant tires for ruggedness and easy maintenance. Chain covers and mudguards can optionally be fitted to improve cleanliness and security. All components are chosen for simplicity in maintenance, accessibility, and affordability. The blend of standard parts with ergonomic additions makes the bicycle extremely functional. Together, these systems all help to provide a safe, low-maintenance, and rider-convenient cycling experience.'

HAND SKETCHES

Freehand sketch of a bicycle:



2D sketch of the Bicycle:



3D Sketch of the Bicycle:



CONCEPT SELECTION & FINALISATION WITH LOGICAL REASONING

Frame Selection:

Following frame type comparisons, the Z-frame design was chosen for its excellent stability, strength, and contemporary looks. Its angular design provides improved load distribution and balance, perfect for riding in urban areas. The design also improves portability and storage, making it appropriate for commuter use. The small geometry is structurally efficient and visually unique in a bicycle frame.

Comfort Optimisation:

Rather than employing a conventional height-adjustable seat, the design features a spring-based system under the seat. The spring suspension, aided by joints, takes in shock and vibrations, providing greater comfort over bumpy roads. Not having adjustable mechanisms minimises complexity and maximises durability. This design option favours rider comfort without adjustable requirements, maintaining simplicity and simplicity.

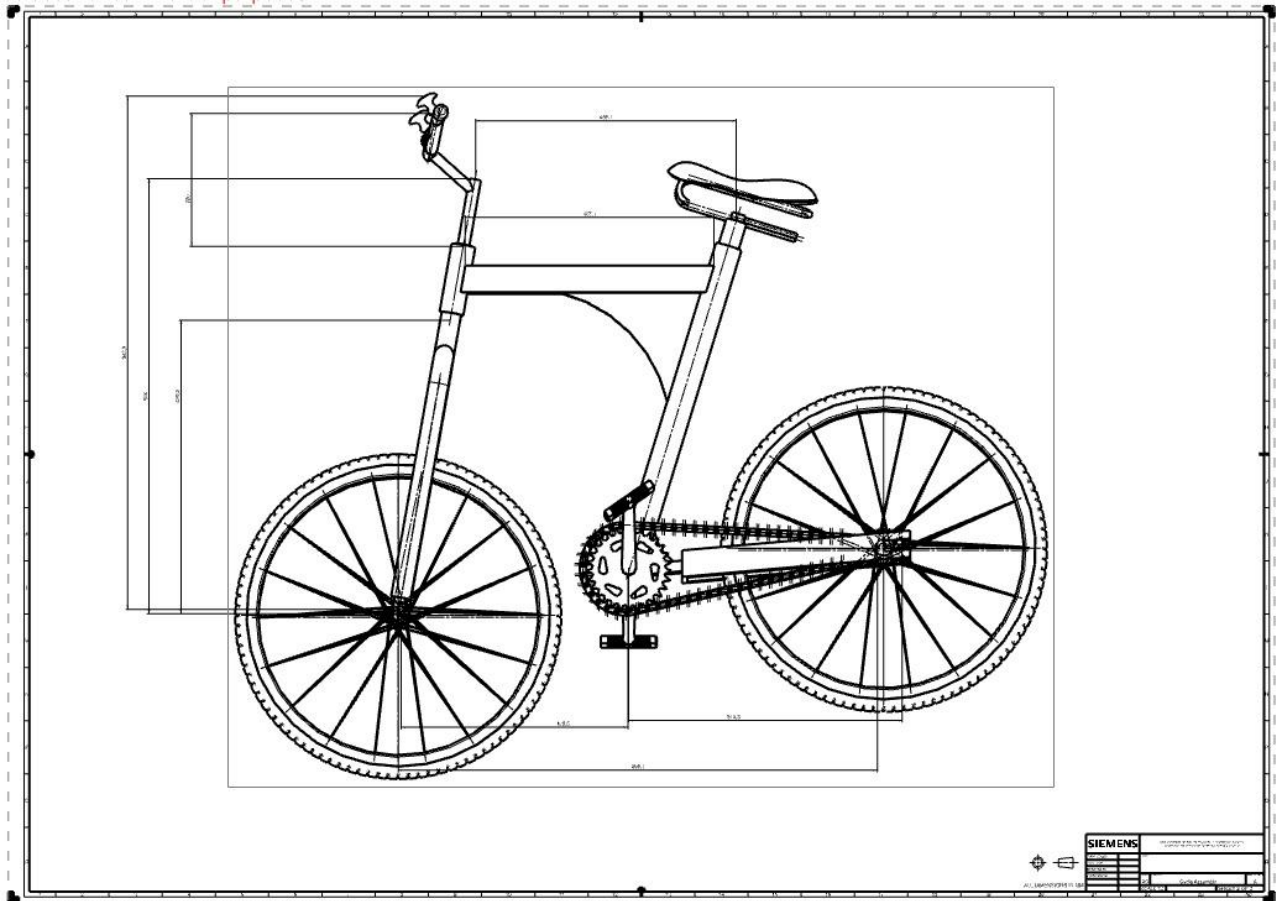
Material Selection:

To guarantee long-term performance, the bicycle frame and key components are constructed using Colled rolled (CR) metals. These metals provide excellent strength, resistance to rust, and long-term endurance in different weather conditions. CR metals also fit sustainable design objectives, as they can be recycled and do not require frequent repairs or replacement.

Component Finalisation:

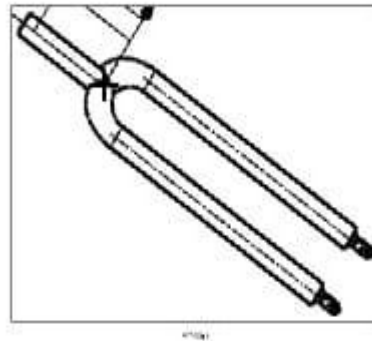
Each component of the bicycle was selected for its practicality, usability, and manufacturability. Platform pedals offer a safe, slip-free surface for any user. The handlebar has improved grip and ergonomic support of the rider's posture to give a better feel for control and less fatigue. A kickstand mounted on the side offers added convenience for rapid and stable parking. Each component facilitates a balanced combination of performance, comfort, and simplicity designed for everyday use.

Cycle assembly Master layout



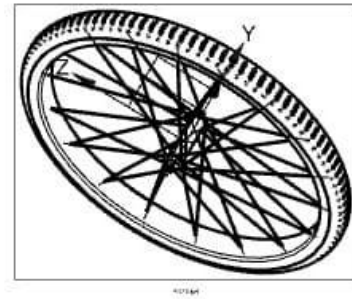
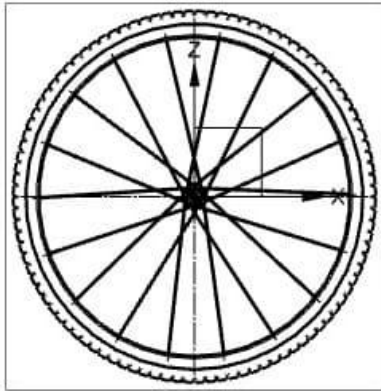
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Team Name	Pegasus
College	Knowledge Institute of Technology

The Frame of a Bicycle



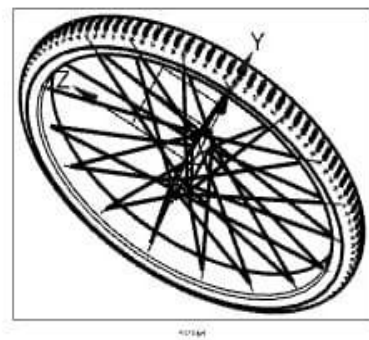
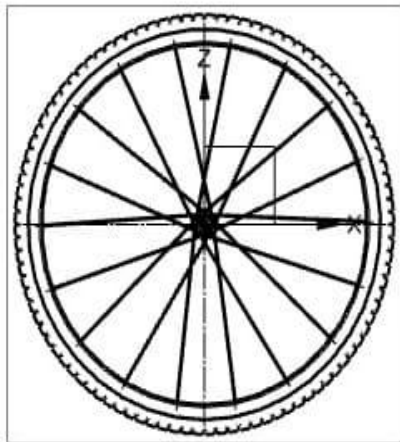
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Pedal of a Bicycle



Scale 1:1	All Dimensions Are In mm
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Tyre of a Bicycle



Scale 1:1	All Dimensions Are In mm
Team Number	BDC2025020
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BICYCLE SPECIFICATION FINALISATION

Component	Specification
Frame	Built with CR (cold rolls) material for high strength and durability.
Frame Type	Z structure for improved balance, weight distribution, and compactness.
Braking System	Standard mechanical brakes for safe, efficient, and low-maintenance stopping.
Suspension	Single spring suspension located below the seat; provides shock absorption and adds stability to the ride.
Seat	Non-adjustable seat with a spring mechanism underneath to ensure a smooth and comfortable riding experience.
Handlebar	Ergonomically designed with enhanced grip and proper posture alignment.
Pedals	Platform-type pedals for wide, stable, and slip-resistant foot placement.

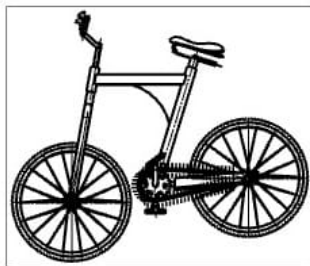
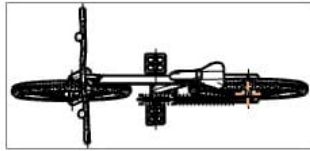
Parking Support	Simple side-mounted kickstand for quick and convenient parking.
Wheels & Tires	Standard wheels with urban tires for reliable traction and smooth road contact.
Material Focus	All components prioritise eco-friendliness, low maintenance, and sustainability.

Unique Component Lists:

- Z Frame Structure
- Seat-mounted spring suspension
- CR (corrosion-resistant) frame material
- Ergonomic handlebar with enhanced grip
- Platform pedals
- Side-mounted kickstand
- Eco-friendly components.

CONVERSION FROM CONCEPT TO MODEL/DRAWING:

Concept To Drawing:



3D View:



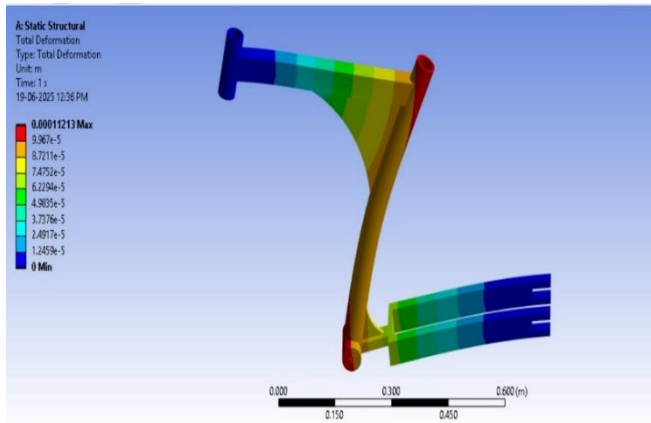
Front View:



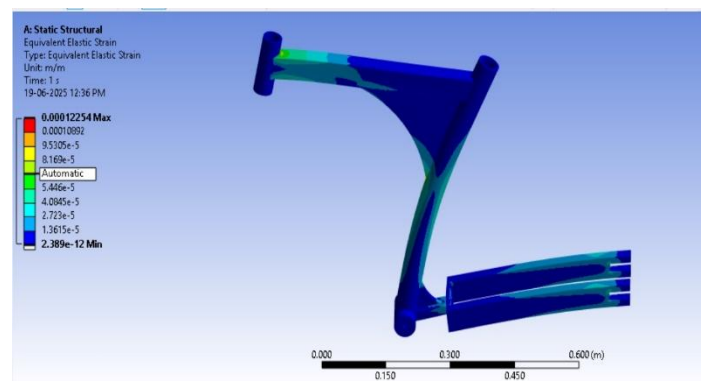
Side View:



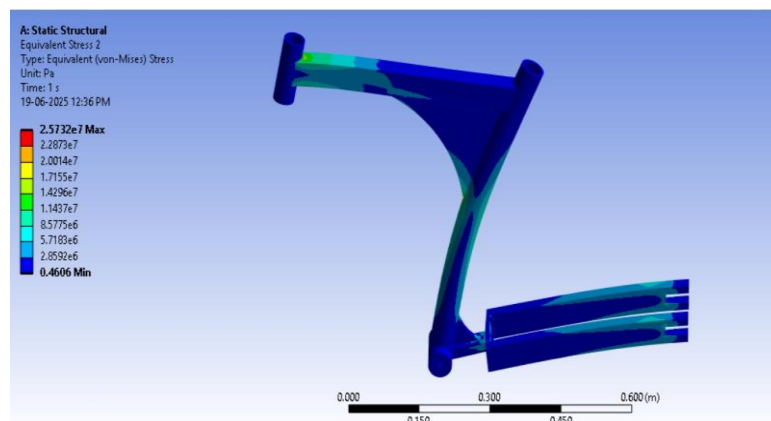
FINITE ELEMENT ANALYSIS:



Total Deformation



Equivalent Elastic strain



Equivalent Stress

BASIC CHECK REQUIREMENTS ON BICYCLE GEOMETRY/GROUND CLEARANCE/TOE CLEARANCE/STEERING GEOMETRY AS PER REGULAR ISO BICYCLE STANDARDS

Bicycle Geometry:

General geometry of the bicycle adheres to conventional ergonomic norms with proper alignment between the handlebars, pedals, and seat for comfort and effectiveness. The Z-frame design is confirmed to be valid for sitting posture, load transfer, and control simplicity.

Head tube angle = 71 degrees

Sear tube angle = 73 degrees

Chain stay length = 17 inches

Ground Clearance:

The distance between the bottom of the frame and the ground is kept constant to prevent bottoming out over obstacles or curbs. The ground clearance is engineered taking into consideration normal urban usage, having sufficient room while also being able to maintain stability when mounting and dismounting.

Wheel radius – Bottom bracket drop

$$13 \text{ in} - 2 \text{ in} = 11 \text{ inch}$$

Toe Clearance:

The design provides adequate space between the front wheel and the toe of the rider under turning conditions. This avoids unintentional contact during steering and aids safe cornering and manoeuvrability, particularly in tight inner-city locations.

Wheel radius – Crank arm length

Wheel radius = 13 inches (for 26-inch wheel)

Typical crank arm length = 6.5 inches

Calculation:

$$\text{Toe clearance} = 13 \text{ in} - 6.5 \text{ in} = 6.5 \text{ inches}$$

Steering Geometry:

The head tube angle and fork rake are tuned to reconcile responsive turning and high-speed stability. The steering mechanism conforms to ISO requirements for smoothness, alignment, and driver control, avoiding wobble or oversteer at low speeds.

Seat To Pedal Distance:

Although the seat is fixed, the fixed geometry and built-in spring design provide a standard ergonomic distance to fit average rider heights comfortably.

Handle Bar Reach and Height:

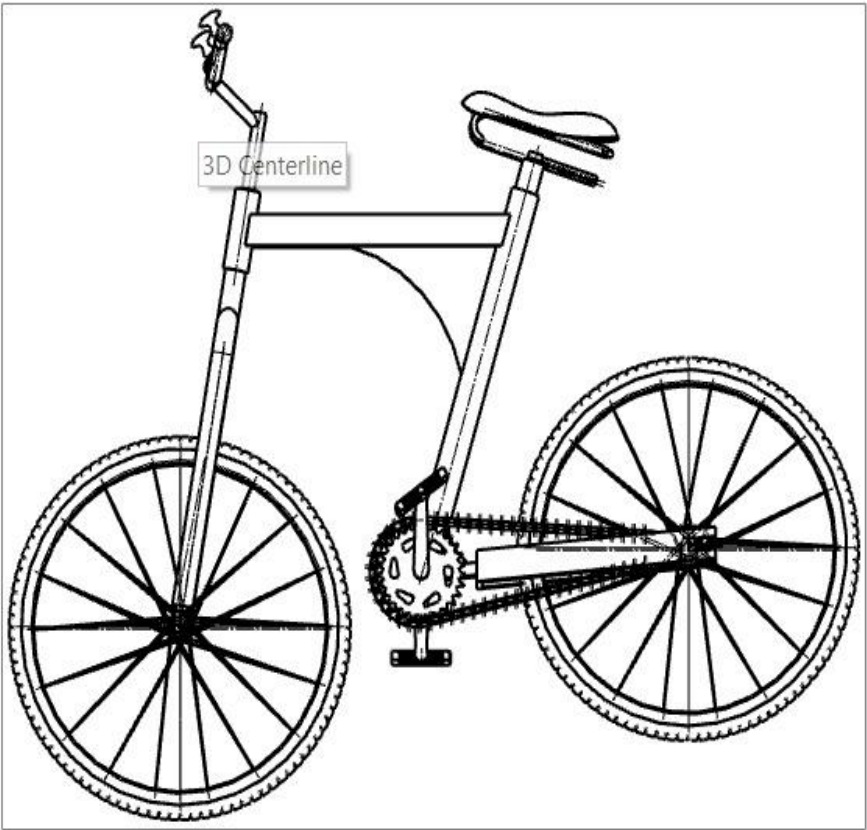
The handlebar position is aligned with ISO-suggested ranges, promoting upright and comfortable riding posture, particularly for city commuting.

ISO Compliance Reference:

All dimensions and ergonomic tolerances are based on ISO 4210 city and trekking bicycle standards, where safety, rider position, and usability are emphasised.

CRITICAL ENGINEERING CALCULATIONS/PRELIMINARY

Detailed part drawing:



NO	DESCRIPTION	QTY
1	HEAD TUBE	1
2	DOWN TUBE	1
3	SEAT TUBE	1
4	TOP TUBE	1
5	REAR TUBE	1
6	REAR WHEEL	1
7	FRONT WHEEL	1
8	PEDESTAL	1
9	CRANK ARM	2
10	PEDAL	2
11	SEAT	1
12	HANDLEBAR	1
13	STAY	1
14	SPACER	1
15	SPACER	1
16	SPACER	1
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DESIGN REVIEW

Team Meeting and Discussion Setup:

Our team had a focused design review meeting to assess the progress and make crucial design decisions. The meeting was attended by the entire team and conducted in an interactive setting, where all members were able to provide feedback and ideas. The meeting was initiated with a presentation of the central design, emphasising the Z-frame structure, seating mechanism, and the choice of materials.

Idea Evaluation and Feedback:

All the design elements were thoroughly discussed. The seat suspension idea (spring under the seat) was argued as an option to height adjustment, and all the members concurred on its simplicity and comfort. Utilisation of CR (corrosion-resistant) material was also favoured for its long lifespan and maintenance benefits. Feedback from members based on user requirements, practicality, and safety was shared.

DISCUSSION, MINUTES OF MEETINGS

Date: 20/06/2025

Attendees: 4 boys, 3 girls

Location : Salem , Tamil Nadu.

Names	Roles
Bhuvanan. M	Captain
Kusumarekha. M	Vice-Captain
Dharaneshwaran. B	Designer-1
Ambika. B	Designer-2
Aravinth. R	Driver
Keerthana. M	Co-Driver
Dharanidharan. T	Research & Development

CONCLUSION

Our team, **Pegasus**, has developed a practical, stable, and eco-friendly bicycle featuring a Z-frame and a spring-based seat suspension for enhanced comfort. Our design ensures durability, user convenience, and aligns with modern urban commuting needs.

